

Sales Data Exploratory Data Analysis and Business Intelligence

Tools Used:

Python, Pandas, Matplotlib, Seaborn, SQLite, SQL, Microsoft Excel

Dataset: ApexPlanet Cleaned Sales Dataset

1. INTRODUCTION:

This project is part of Task 2 of the ApexPlanet Data Analytics Internship. The objective of this task is to perform Exploratory Data Analysis (EDA) and Business Intelligence analysis on the cleaned sales dataset.

The analysis focuses on understanding sales patterns, customer characteristics, product performance, category performance, geographical performance, time-based trends, and relationships between numerical variables.

The analysis uses Python for data exploration and visualization, SQL for answering business questions, and Excel for creating a static business dashboard.

The official Task 2 focuses on uncovering patterns, trends, and relationships in the data while developing proficiency in SQL and basic dashboarding.

2. OBJECTIVES:

The main objectives of this project are:

1. To understand the structure and characteristics of the cleaned dataset.
2. To calculate descriptive statistics for numerical variables.
3. To analyze categorical variables using frequency distributions.
4. To create visualizations to understand data distributions.
5. To formulate and answer business questions using SQL.
6. To analyze relationships between numerical variables.
7. To perform correlation analysis.
8. To identify important business insights from the dataset.

9. To create a static dashboard containing important business KPIs.
10. To present the findings in a clear and business-oriented manner.

1. DATASET OVERVIEW:

The dataset contains sales transaction information involving customers, products, categories, locations, quantities, prices, and sales amounts.

The cleaned dataset contains the following major fields:

Column	Description
Transaction_ID	Unique identifier for a transaction
Order_ID	Identifier for an order
Order_Date	Date on which the order was placed
Customer_ID	Unique customer identifier
Customer_Name	Name of the customer
Age	Age of the customer
Gender	Gender of the customer
City	Customer/order city
Product	Product purchased
Category	Product category
Quantity	Quantity purchased
Unit_Price	Price per unit
Total_Sales	Total sales value
Order_Month	Numerical month of the order
Order_Year	Year of the order
Order_Month_Name	Name of the order month

4. DATASET SIZE:

After loading the cleaned dataset into Python, the dataset was examined using Pandas.

Dataset dimensions

Rows: 1,000

Columns: 16

Therefore, the dataset contains **1,000 transaction records and 16 variables**.

The cleaned dataset was used directly for the EDA without repeating the cleaning process performed during Task 1.

5. TOOLS AND TECHNOLOGIES:

The following tools were used:

Python

Python was used as the primary programming language for data analysis.

Pandas

Pandas was used for:

- Loading the Excel dataset
- Data inspection
- Descriptive statistics
- Grouping and aggregation
- Data analysis

Matplotlib

Matplotlib was used to create:

- Histograms
- Bar charts
- Line charts
- Scatter plots

Seaborn

Seaborn was used primarily for the correlation heatmap.

SQLite

SQLite was used to create a local SQL database from the cleaned dataset.

SQL

SQL was used to answer business questions involving filtering, grouping, aggregation, and sorting.

Microsoft Excel

Excel was used to create the static dashboard mock-up.

6. DESCRIPTIVE STATISTICS:

Descriptive statistics were calculated using the describe() function in Pandas.

The main numerical variables analyzed were:

- Age
- Quantity
- Unit_Price
- Total_Sales

The descriptive statistics include:

- Count
- Mean
- Standard deviation
- Minimum
- 25th percentile
- Median
- 75th percentile
- Maximum

Descriptive Statistics:

	Order_Date	Age	Quantity	Unit_Price \
count	1000	1000.000000	1000.000000	1000.000000
mean	2025-07-03 11:19:40.800000	41.353000	5.435000	25486.783410
min	2025-01-01 00:00:00	18.000000	1.000000	145.780000
25%	2025-03-30 18:00:00	30.000000	3.000000	13895.722500
50%	2025-07-01 00:00:00	41.000000	5.000000	25398.740000
75%	2025-10-07 06:00:00	53.000000	8.000000	37512.382500
max	2026-01-01 00:00:00	65.000000	10.000000	49997.530000
std	NaN	13.683626	2.838632	14179.402361

	Total_Sales	Order_Month	Order_Year
count	1000.000000	1000.000000	1000.000000
mean	139399.439650	6.518000	2025.005000
min	437.340000	1.000000	2025.000000
25%	47066.632500	3.000000	2025.000000
50%	108594.025000	6.000000	2025.000000
75%	203722.882500	10.000000	2025.000000
max	493677.500000	12.000000	2026.000000
std	114100.051546	3.498845	0.070569

7. KEY BUSINESS METRICS:

Several important business metrics were calculated from the dataset.

TOTAL SALES	TOTAL ORDERS	TOTAL CUSTOMERS	TOTAL QUANTITY	AVERAGE TRANSACTION SALES
₹ 139,399,439.65	994	949	5435	139399.4397

These metrics provide a high-level overview of the sales performance represented in the dataset.

8. CATEGORICAL DATA ANALYSIS:

Categorical variables were analyzed using frequency counts.

The main categorical variables considered were:

- Gender

- City
- Product
- Category

The `value_counts()` function was used to understand the frequency of different categories.

9. GENDER DISTRIBUTION:

The gender distribution was analyzed to understand the number of transactions associated with male and female customers.

```
df["Gender"].value_counts()
```

Interpretation

The analysis shows the distribution of transactions across the two gender groups. This helps understand whether the transaction dataset is relatively balanced between male and female customers.



10. PRODUCT DISTRIBUTION:

The dataset contains six major products:

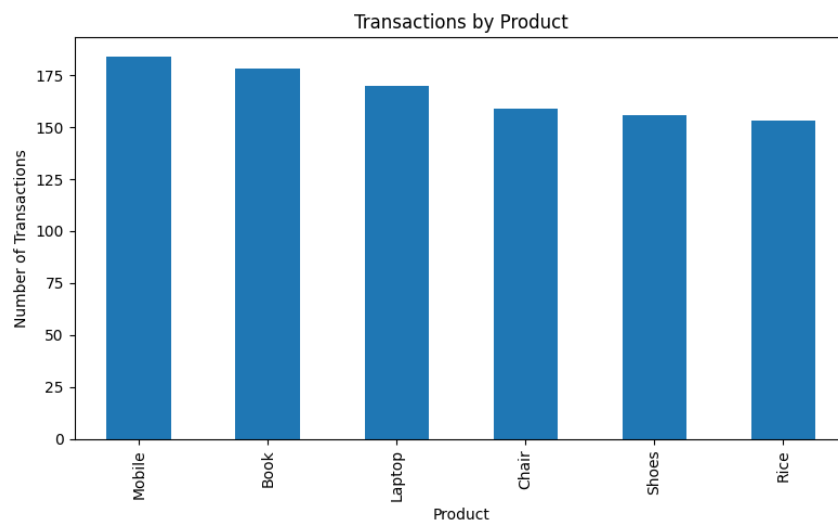
- Rice

- Book
- Mobile
- Laptop
- Shoes
- Chair

The frequency of transactions for each product was analyzed.

Interpretation

Product-level frequency analysis helps identify which products appear most frequently in the transaction records.



11. CATEGORY DISTRIBUTION:

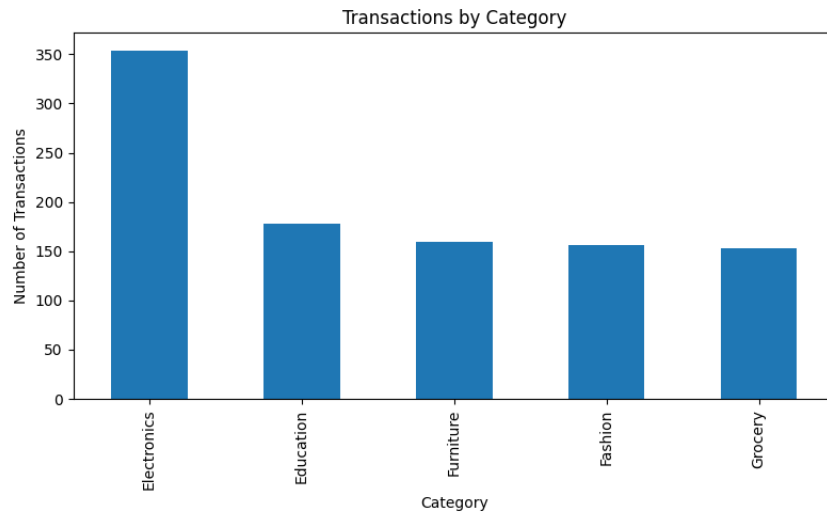
The dataset contains the following major product categories:

- Grocery
- Education
- Electronics
- Fashion
- Furniture

The distribution of transactions across categories was analyzed.

Interpretation

Category-level analysis helps identify the relative representation of different product categories in the dataset.



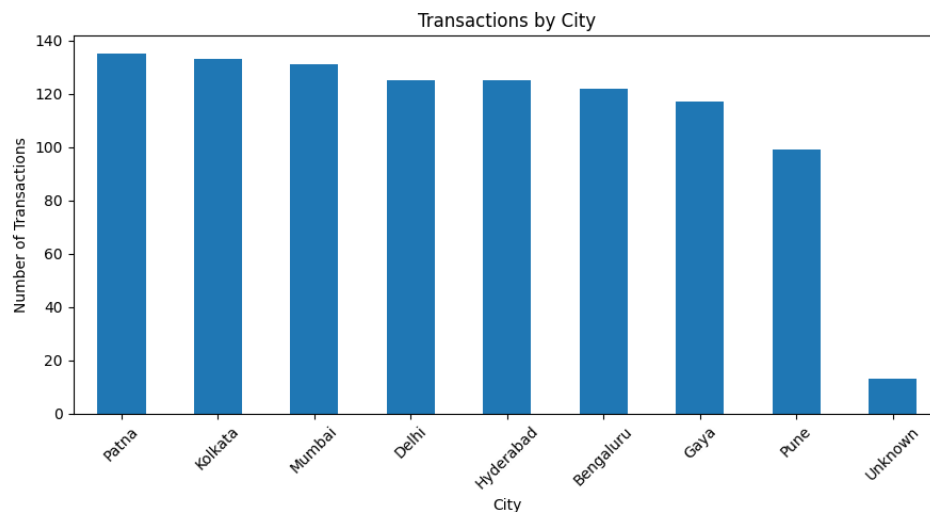
12. CITY DISTRIBUTION:

The dataset contains transactions associated with multiple cities.

The city distribution was analyzed using a bar chart.

Interpretation

This analysis helps identify the cities that contribute the largest number of transaction records and provides an overview of the geographical distribution of the dataset.



13. UNIVARIATE ANALYSIS:

Univariate analysis examines one variable at a time.

The following variables were analyzed:

- Age
- Quantity
- Unit Price
- Total Sales
- Gender
- Product
- Category
- City

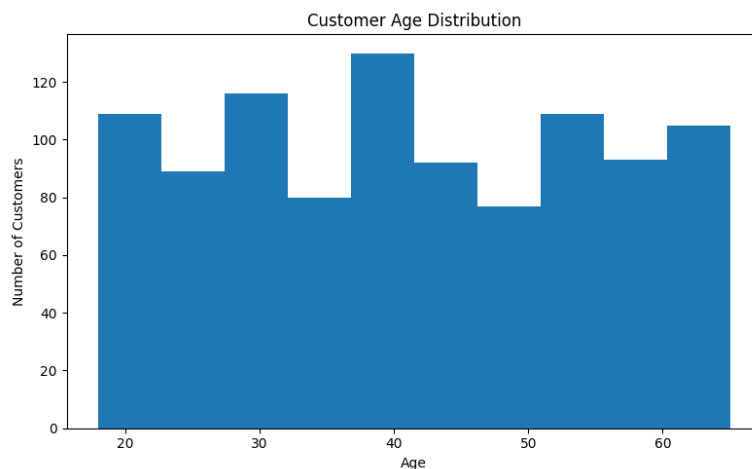
Histograms were used for numerical variables, while bar charts were used for categorical variables.

14. CUSTOMER AGE DISTRIBUTION:

A histogram was created to understand the distribution of customer ages.

Interpretation

The age distribution provides an overview of the customer demographic represented in the dataset. It can help identify the major age ranges present among customers.

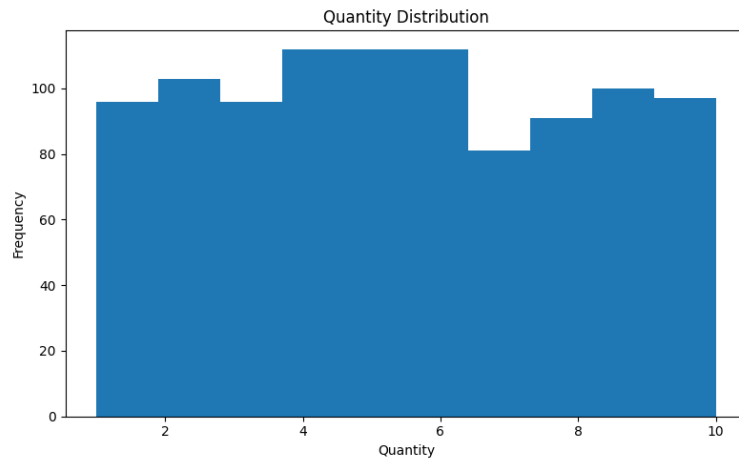


15. QUANTITY DISTRIBUTION:

A histogram was created for the Quantity variable.

Interpretation

The quantity distribution shows how many units customers typically purchase per transaction. This can help identify common purchasing quantities and unusually high quantities.

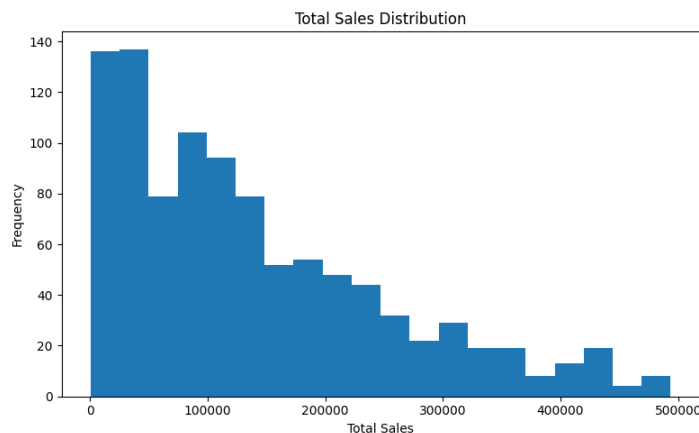


16. TOTAL SALES DISTRIBUTION:

The distribution of Total_Sales was visualized using a histogram.

Interpretation

The sales distribution helps understand the range and frequency of transaction values. It can also help identify whether most transactions have relatively small values or whether some transactions have considerably larger sales values.



17. SQL BUSINESS ANALYSIS:

SQL was used to answer seven business questions based on the cleaned sales dataset.

The SQL analysis uses aggregation, grouping, sorting, and average calculations.

The internship specifically requires 5–7 specific business questions and SQL queries with results.

18. BUSINESS QUESTION 1:

Which products generate the highest total sales?

```
SELECT Product,  
        SUM(Total_Sales) AS Total_Sales  
FROM sales  
GROUP BY Product  
ORDER BY Total_Sales DESC;
```

Purpose

This query identifies the products that contribute the highest amount of total sales.

Product	Total_Sales
Laptop	25443009
Mobile	25335573
Book	25031689
Rice	22231711
Chair	21521561
Shoes	19835896

Business Interpretation

The result ranks products according to their total sales contribution. The highest-ranked product represents the strongest revenue-generating product in the dataset.

19. BUSINESS QUESTION 2:

Which product categories generate the highest total sales?

```
SELECT Category,  
        SUM(Total_Sales) AS Total_Sales
```

FROM sales

GROUP BY Category

ORDER BY Total_Sales DESC;

Category	Total_Sales
Electronics	50778582
Education	25031689
Grocery	22231711
Furniture	21521561
Fashion	19835896

Business Interpretation

The query identifies the categories contributing the most to overall sales. This information can help businesses prioritize high-performing product categories.

20. BUSINESS QUESTION 3:

Which cities generate the highest total sales?

SELECT City,

SUM(Total_Sales) AS Total_Sales

FROM sales

GROUP BY City

ORDER BY Total_Sales DESC;

City	Total_Sales
Patna	19285967
Kolkata	18884350
Bengaluru	18773574
Mumbai	18757050
Hyderabad	17166767
Delhi	16097079
Pune	14513176
Gaya	14380859
Unknown	1540618

Business Interpretation

The result shows sales performance by city. Higher-performing cities may represent stronger markets and can be considered for future sales and marketing strategies.

21. BUSINESS QUESTION 4:

Which products have the highest quantity sold?

```
SELECT Product,  
        SUM(Quantity) AS Total_Quantity  
FROM sales  
  
GROUP BY Product  
  
ORDER BY Total_Quantity DESC;
```

Product	Total_Quantity
Mobile	1008
Book	977
Laptop	970
Chair	855
Rice	826
Shoes	799

Business Interpretation

This analysis identifies the products with the highest sales volume. A product with high quantity sold may have strong customer demand even if its unit price is relatively low.

22. BUSINESS QUESTION 5

What is the monthly sales trend?

```
SELECT Order_Year,  
        Order_Month,  
        Order_Month_Name,  
        SUM(Total_Sales) AS Total_Sales  
FROM sales  
  
GROUP BY Order_Year, Order_Month, Order_Month_Name  
  
ORDER BY Order_Year, Order_Month;
```

Order_Year	Order_Month	Order_Month_Name	Total_Sales
2025	1	January	10096191
2025	2	February	11511181

2025	3	March	13059900
2025	4	April	12222700
2025	5	May	10984689
2025	6	June	12912333
2025	7	July	11746227
2025	8	August	9448471
2025	9	September	9179896
2025	10	October	12500937
2025	11	November	12627620
2025	12	December	12299904
2026	1	January	809390

Business Interpretation

The monthly sales analysis helps identify changes in sales over time. It can be used to identify stronger and weaker sales periods.



23. BUSINESS QUESTION 6:

How do total sales compare between male and female customers?

SELECT Gender,

SUM(Total_Sales) AS Total_Sales

FROM sales

GROUP BY Gender

ORDER BY Total_Sales DESC;

Gender	Total_Sales
Male	72463551
Female	66935889

Business Interpretation

The result compares the contribution of male and female customers to overall sales. This can provide useful information for customer segmentation and marketing decisions.

24. BUSINESS QUESTION 7:

Which products have the highest average unit price?

```
SELECT Product,  
  
       AVG(Unit_Price) AS Average_Unit_Price  
  
FROM sales  
  
GROUP BY Product  
  
ORDER BY Average_Unit_Price DESC;
```

Product	Average_Unit_Price
Rice	26360.24
Shoes	25895.97
Chair	25788.9
Laptop	25638.27
Book	24746.71
Mobile	24728.48

Business Interpretation

This query identifies products with the highest average selling price. Comparing average price with quantity sold can help understand product positioning and customer purchasing behavior.

25. MULTIVARIATE ANALYSIS:

Multivariate analysis was performed to understand relationships between multiple numerical variables.

The main variables considered were:

- Age
- Quantity
- Unit_Price
- Total_Sales

Scatter plots and correlation analysis were used.

The internship specifically recommends advanced visualizations such as scatter plots, heatmaps, and pair plots for this stage.

26. CORRELATION ANALYSIS:

A correlation matrix was calculated using:

```
correlation = df[  
    ["Age", "Quantity", "Unit_Price", "Total_Sales"]  
].corr()
```

The correlation coefficient ranges from:

-1 to +1

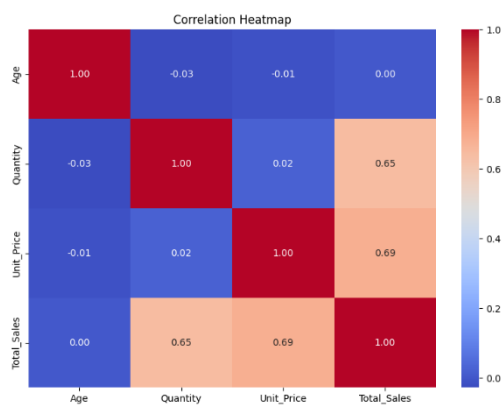
where:

- +1 indicates a strong positive relationship
- 0 indicates little or no linear relationship
- -1 indicates a strong negative relationship

	Age	Quantity	Unit_Price	Total_Sales
Age	1.000000	-0.027666	-0.011955	0.001294
Quantity	-0.027666	1.000000	0.021855	0.646641
Unit_Price	-0.011955	0.021855	1.000000	0.686303
Total_Sales	0.001294	0.646641	0.686303	1.000000

27. CORRELATION HEATMAP:

A heatmap was created to visually represent the correlation between numerical variables.



Interpretation

The heatmap makes it easier to identify variables that have relatively stronger positive or negative linear relationships.

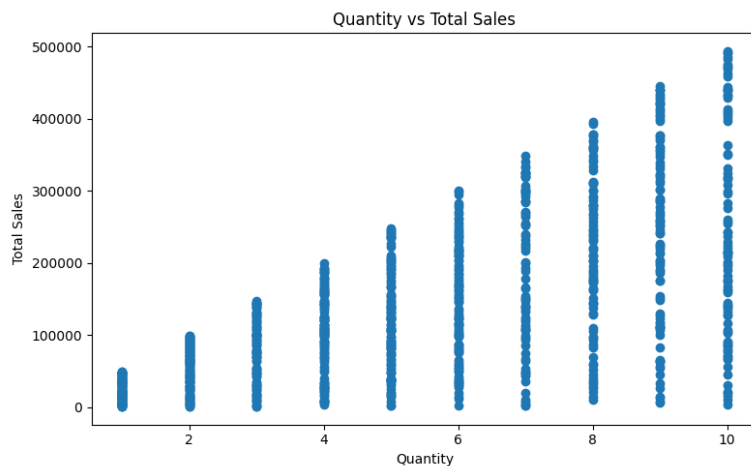
Particular attention was given to the relationship between:

- Quantity and Total Sales
- Unit Price and Total Sales
- Age and Total Sales
- Quantity and Unit Price

The exact interpretation should be based on the correlation coefficients obtained from the analysis.

28. QUANTITY VS TOTAL SALES:

A scatter plot was created to investigate the relationship between quantity purchased and total sales.



Interpretation

This visualization helps determine whether transactions involving larger quantities tend to have higher total sales values.

29. UNIT PRICE VS TOTAL SALES:

A scatter plot was created between unit price and total sales.

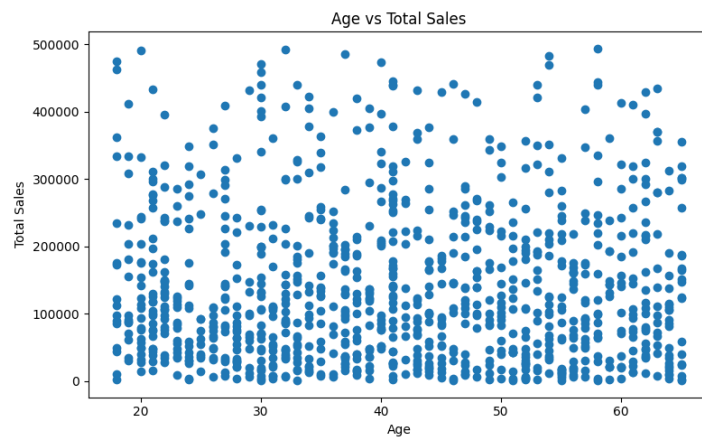


Interpretation

This analysis helps examine whether higher-priced products or transactions are associated with larger total sales values.

30. AGE VS TOTAL SALES:

A scatter plot was created to examine the relationship between customer age and total sales.



Interpretation

This visualization helps determine whether customer age appears to have a noticeable linear relationship with transaction sales.

31. KEY BUSINESS INSIGHTS:

Based on the EDA and SQL analysis, the following areas were examined:

Product Performance

The products were compared based on total sales and quantity sold. This provides information about both revenue performance and sales volume.

Category Performance

Product categories were compared to identify the categories contributing the greatest sales.

Geographical Performance

Sales were analyzed by city to identify stronger and weaker geographical markets.

Customer Analysis

Gender and age were analyzed to understand customer characteristics.

Time-Based Performance

Monthly sales were analyzed to identify changes in sales performance over time.

Relationship Analysis

Correlation and scatter plots were used to examine relationships among age, quantity, unit price, and total sales.

32. DASHBOARD:

A static sales analytics dashboard was created in Microsoft Excel based on the results of the EDA.

The dashboard focuses on important business KPIs and visual summaries.

The dashboard contains:

KPI 1 — Total Sales

Shows the overall sales value.

KPI 2 — Total Orders

Shows the number of unique orders.

KPI 3 — Total Customers

Shows the number of unique customers.

KPI 4 — Total Quantity

Shows the total number of units sold.

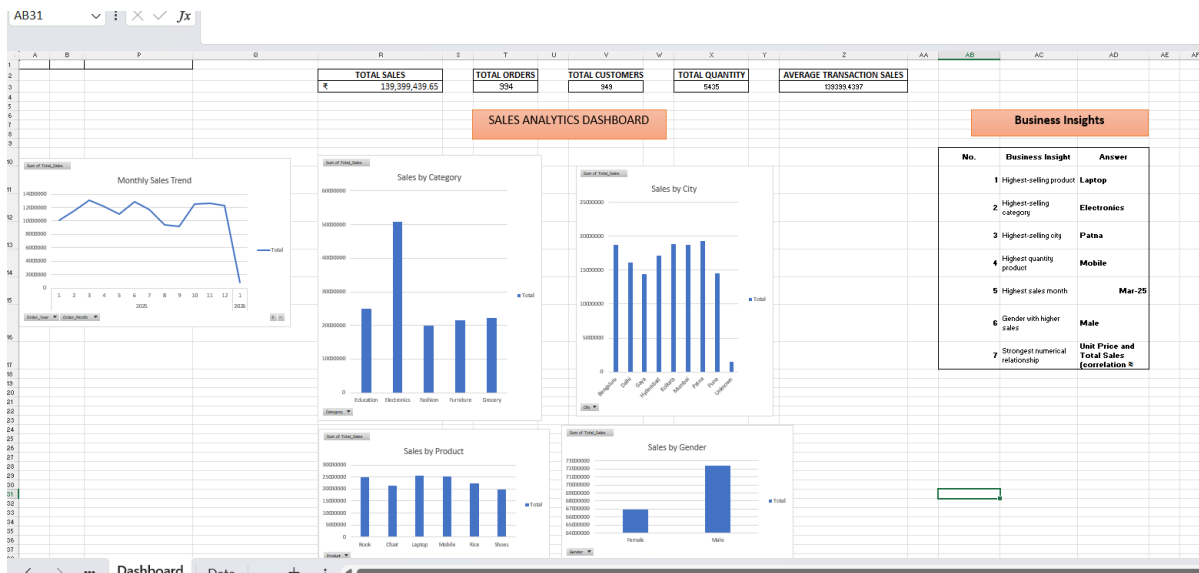
KPI 5 — Average Transaction Sales

Shows the average sales value per transaction.

The dashboard also contains visualizations for:

- Sales by Category
- Sales by Product
- Sales by City
- Sales by Gender
- Monthly Sales Trend

The internship specifically asks for a static dashboard mock-up in Excel, PowerPoint, or Google Slides based on EDA insights.



33. CONCLUSION:

The Exploratory Data Analysis provided a comprehensive understanding of the sales dataset by examining numerical distributions, categorical variables, sales performance, customer characteristics, geographical patterns, and time-based trends.

The SQL analysis helped answer important business questions related to products, categories, cities, sales quantity, monthly performance, gender, and product pricing.

Multivariate analysis and correlation analysis were used to examine relationships between numerical variables, while scatter plots and a correlation heatmap provided visual insights into these relationships.

Finally, the findings were incorporated into a static dashboard containing key sales KPIs and business-oriented visualizations.

Overall, this project demonstrates the application of Python, SQL, data visualization, statistical analysis, and dashboard design for extracting meaningful business insights from transactional data.

34. RECOMMENDATIONS:

Based on the analysis framework, the following recommendations can be considered:

1. Focus on products that demonstrate strong total sales and consistent demand.
2. Monitor high-performing categories and consider prioritizing them in future sales strategies.
3. Analyze high-performing cities to understand the factors contributing to stronger sales.
4. Track monthly sales trends regularly to identify changes in demand.
5. Use customer demographic information to support targeted marketing strategies.
6. Compare product quantity and unit price to understand the relationship between sales volume and revenue.
7. Continue monitoring the dashboard KPIs to support data-driven business decisions.

